🇦🇹 Austria — Substation ESG Report v4.2

Substation-level Environmental, Social, and Governance assessment aligned to CSRD/ESRS, EU Taxonomy, TCFD, SFDR, and UN PRI frameworks, across the 14,720-substation Austria fleet. Updated monthly via the SSI Index v4.2 scoring engine (101 variables · 34 sources · 11 modifiers · Re composite bounded [0.920, 1.787]).

SUBSTATION

Station v43_80c76c1cf90e

AT_v43_80c76c1cf90e · , Traunviertel

SSI SCORE (R_{MEDIAN})

0.101

● Low Risk 21.6th percentile

VOLTAGE CLASS

30 kV

distribution-tier

CONFIDENCE INTERVAL

0.101 – 0.101

P5–P95 · CI width 0.000 · confidence

MARKOV ETTC

— yr

Expected time to criticality

DER PENETRATION

—

T1 transition stress score

Component Fingerprint + Re composite v4.2

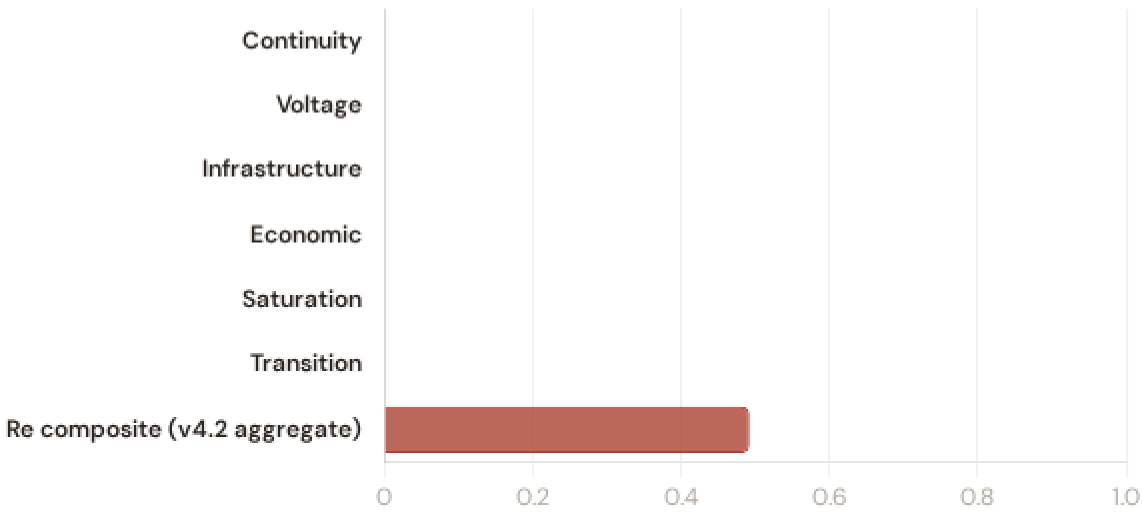
Six-component weighted decomposition of the SSI composite score (C / V / I / E / S / T) — the substation's structural risk profile — PLUS the v4.2 **Re composite**, a bounded resilience aggregate of the new modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just). Layout: *left* — the 6 components rendered as a weighted contribution bar · *right* — the same 6 components rendered as horizontal bars side-by-side with Re_norm as the comparable 7th bar (terracotta) on the [0, 1] scale · *below* — the Re composite tier card (numeric value + bounded trajectory bar + classification + formula explainer). Re composite re-appears as the 7th axis of the ESG Radar in Section A below.

WEIGHTED CONTRIBUTION TO R_{BASE}

V4.2 RE COMPOSITE (AGGREGATE, NOT IN R_{BASE}) moderate exposure

- Continuity — ×0.3
- Voltage — ×0.1
- Infrastructure — ×0.25
- Economic — ×0.1
- Saturation — ×0.2
- Transition — ×0.05
- **Re composite** v4.2 aggregate **0.493** agg

R_{base} **total** (weighted sum) **0.000** Σ = 1.00



0.493

RE_NORM

MODERATE EXPOSURE

LOW (0.00) MODERATE (0.25) ELEVATED (0.50) (1.00)

Re_raw 1.3470 (bounds [0.920, 1.787]; normalised to [0, 1])

FORMULA

```
Re_raw = (R6d × R6e × R8 × R9 ×
R10) + (R6c − 1.00)
```

Bounded resilience aggregate of the v4.2 family. Re < 1.0 when R8 adaptive capacity dominates other near-neutral factors.

Re composite tier classification: **low** (Re_norm < 0.25) · **moderate** (0.25–0.50) · **elevated** (≥ 0.50). Bounds [0.920, 1.787] map to Re_norm [0, 1] via clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1).

Markov Degradation Profile

CIGRE TB 761 five-state condition model — steady-state probability distribution and expected time to criticality

STEADY-STATE DISTRIBUTION

GOOD	AGED	DEGRADED	CRITICAL
0%	0%	0%	0%
$\pi 1$ = 0.000	$\pi 2$ = 0.000	$\pi 3$ = 0.000	$\pi 4$ = 0.000

Markov 5-state model calibrated from CIGRE Technical Brochure 761 (2019). States represent asset condition from new/good through critical. Steady-state probabilities indicate the long-run proportion of time the asset spends in each condition state given current operating environment.

RISK SCORE

—

Composite Markov risk

ETTC

—

Expected time to critical

P(CRITICAL) 20YR

—

Probability of reaching critical

CORROSION

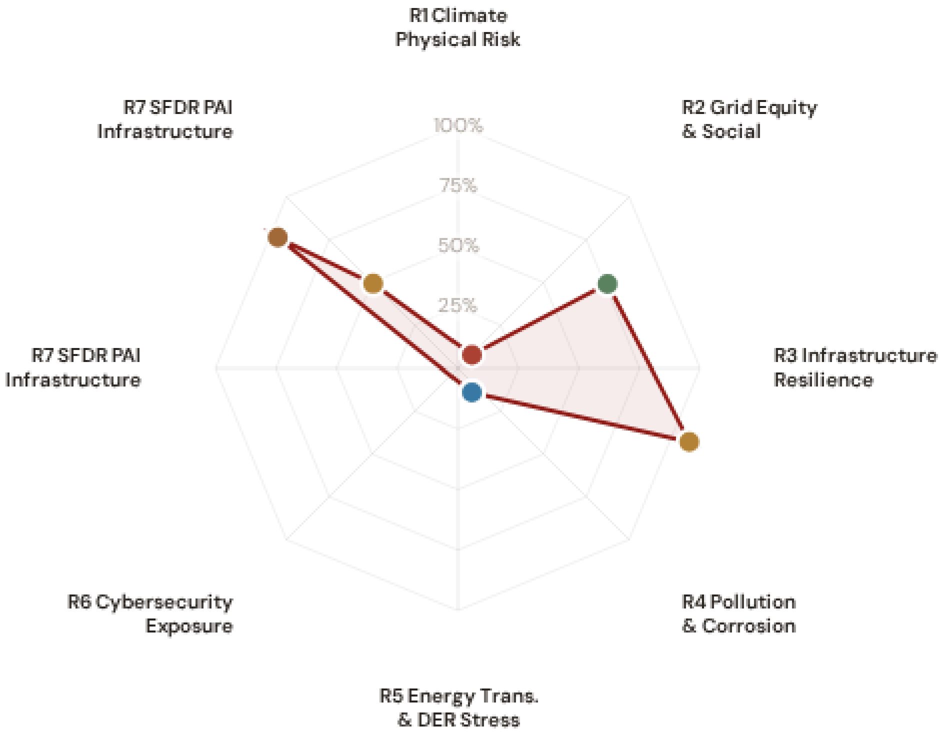
—

ISO 9223 classification

ESG Radar — Seven-Report Overview V4.2

A

Composite readiness across 7 ESG dimensions per FC v3 §14 (R1 Climate Physical · R2 Grid Equity & Social · R3 Infrastructure Resilience [Re composite ESG home] · R4 Pollution & Corrosion · R5 Energy Transition & DER Stress · R6 Cybersecurity · R7 SFDR PAI Infrastructure, FC v3 §14 subsection 13.7), each linked to UN Sustainable Development Goals. *Note: R7 here is the SFDR ESG-disclosure axis (Re_norm proxy under Convention #7) — distinct from the v4.0.2 formula-chain R7 cyber/digital modifier documented in methodology.html. The two R7s share a number by design (V4_2_IMPLEMENTATION_ARCHITECTURE.md §4.5).*



Re composite v4.2 7th axis · **Re_raw** 1.3470 · **Re_norm** 0.493 · MODERATE EXPOSURE

- **R1 · Climate Physical Risk Assessment** GAP
  SDG 13 Climate Action · SDG 9 · SDG 11
  SDG 13 SDG 9 SDG 11

- **R2 · Grid Equity & Social Vulnerability** PARTIAL
  SDG 7 Affordable and Clean Energy · SDG 10 · SDG 1
  SDG 7 SDG 10 SDG 1

- **R3 · Infrastructure Resilience [Re composite home]** READY
  SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 12
  SDG 9 SDG 13 SDG 12

- **R4 · Pollution & Corrosion** GAP
  SDG 11 Sustainable Cities and Communities · SDG 3 · SDG 15
  SDG 11 SDG 3 SDG 15

- **R5 · Energy Transition & DER Stress** GAP
  SDG 7 Affordable and Clean Energy · SDG 13 · SDG 9

SDG 7 SDG 13 SDG 9

R6 · Cybersecurity Exposure GAP

SDG 9 Industry, Innovation, Infrastructure · SDG 16

SDG 9 SDG 16

R7 · SFDR PAI Infrastructure Disclosure READY

SDG 12 Responsible Consumption and Production · SDG 9 · SDG 13

SDG 12 SDG 9 SDG 13

R7 · SFDR PAI Infrastructure DEFAULT v4.2

SDG 9 Industry, Innovation, Infrastructure · SDG 13 · SDG 11 · Re_norm = 0.493 (moderate exposure)

SDG 9 SDG 13 SDG 11

1

Climate Physical Risk Assessment

ESRS E1 · TCFD Physical Risk · EU Taxonomy Annex A

GAP

WHY THIS REPORT EXISTS

ESRS E1 (Climate Change) is material for 98% of CSRD-reporting companies. TCFD physical risk disclosure is mandatory in 36+ jurisdictions. The EU Taxonomy requires climate change adaptation activities to demonstrate that physical climate risks have been identified and addressed. This report quantifies acute and chronic climate hazard exposure at substation level using ERA5 reanalysis and Markov degradation modelling.

SDG Alignment

SDG 13 — Climate Action (primary)

Target 13.1 calls for strengthening resilience and adaptive capacity to climate-related hazards. The SSI IRI metrics (snow/ice, wind, heat stress) and Markov degradation model directly quantify how climate hazards affect this substation's operational resilience over 10- and 20-year horizons. With a Markov risk score of — and ETTC of — years, this substation's climate trajectory can be assessed against TCFD physical risk and EU Taxonomy adaptation screening requirements.

SDG 13 SDG 9 SDG 11

Substation Profile

Insulation Risk (I component)	—
Environment Component (E)	—
Seismic PGA	—g (Zone —)
Markov Risk Score	—
ETTC (time to criticality)	— years
P(critical) at 20 years	—
Corrosion Class	—
Climate Trajectory (R2)	NOT ACTIVE

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
I1 Snow/Ice IRI	0.985	ESRS E1-9: Financial effects from physical risks	READY
I2 Tree-fall IRI	1.050	TCFD Strategy (b): Physical risk impact	READY
I3 Heat-wave IRI	1.022	EU Taxonomy Annex A: Heat stress	READY
I5 Thermal stress (I aggregate)	—	ESRS E1-4: Climate mitigation targets	GAP
R2 Climate trajectory	N/A	TCFD Strategy (c): Scenario analysis	GAP
R6b Seismic PGA	—g	EU Taxonomy Annex A: Geophysical hazards	GAP
Markov p_critical_10yr	—	TCFD Risk Management: Risk identification	READY
Markov p_critical_20yr	—	ESRS E1-9: Time horizons	READY

Data Sources & Vintage

ERA5 Climate Reanalysis Copernicus CDS	**2024** Weekly
National Seismic Hazard Map National Geological Survey	**2023** Multi-year
IEEE C57.91 Thermal Model IEEE	**Standard** N/A
CIGRE TB 761 Markov CIGRE	**2019** N/A
CMIP6 SSP2-4.5 Projections Copernicus CDS	**2024** v4.2 5-model ensemble

TECHNICAL VALIDITY

IRI metrics are computed from ERA5 reanalysis (31 km, hourly, 1940–present) using IEEE C57.91 thermal loading curves. Markov 5-state degradation is calibrated from CIGRE Technical Brochure 761 (2019) condition assessment data. Forward-looking projections use the CMIP6 SSP2-4.5 5-model ensemble (ACCESS-CM2, CNRM-CM6-1, EC-Earth3, GFDL-ESM4, MRI-ESM2-0), baseline 2000–2020 vs future 2030–2050, feeding the substation-level climate_trajectory (I1/I2/I3) fields for 32 of 39 countries. Any PARTIAL status now reflects country-specific gaps: 7 Wave 2-3 countries (Australia, Chile, Colombia, Costa Rica, Iceland, Ireland, Israel) await the next L2 climate pass; 8 Wave 4 countries emit fleet-uniform trajectories pending the per-substation bilinear interpolation regression fix (methodology follow-on, sister to R3_C_mult uniformity). All other inputs are production-grade with institutional provenance.

2 Grid Equity & Social Vulnerability

ESRS S1/S2 · SFDR PAI Social Indicators · UN PRI

PARTIAL

WHY THIS REPORT EXISTS

ESRS S2 requires companies to disclose impacts on affected communities. SFDR mandates 5 social PAI indicators. The SSI Index is uniquely positioned here because no competing grid risk framework integrates socio-economic vulnerability at substation resolution. This report surfaces where infrastructure risk concentrates in economically disadvantaged populations.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.1 requires universal access to affordable, reliable energy. This substation serves a catchment with V_socio of 0.38 and energy poverty rate of 10%. The R3 social modifier (—) amplifies risk scores in regions with high unemployment, elderly concentration, and net outward migration — making spatial inequality visible and quantifiable at asset level.

SDG 7 SDG 10 SDG 1

Substation Profile

V_socio (energy poverty index)	**0.38**
EP Rate (energy poverty %)	**10%**
Unemployment Rate	**4.9%**
GDP per Capita	**€ 32,300**
R&D % of GDP	**2.1%**
R3 Social Multiplier	**—**
Elderly Vulnerability	**18.5%**
Migration Score	**0.61**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
V_socio Energy poverty index	0.38	ESRS S2-4: Impacts on affected communities	READY
EP_rate Energy poverty rate	10%	SFDR PAI #14: Fossil fuel exposure proxy	READY
R3 Social multiplier	—	ESRS S1-6: Workforce characteristics	GAP
Elderly Elderly vulnerability	18.5%	SFDR PAI: Vulnerable populations	READY
Community Catchment population	30,751	ESRS S2: Community impact	READY

Data Sources & Vintage

GHSL Population Grid
EC JRC / Copernicus
2025 Multi-year

Global Net-Migration Grid
Niva et al. 2023 (Aalto / Wittgenstein)
2000-2019 Multi-year

Eurostat GISCO NUTS-3 Regions
European Commission / Eurostat
2024 Annual

GADM Administrative Areas 4.1 (CO / CH / CA)
UC Davis geodata.ucdavis.edu/gadm
2022 Multi-year

Economic Statistics (GDP, unemployment)
National Statistics Office
2023 Annual

Energy Market Data
National TSO
2023 Annual

TECHNICAL VALIDITY

V_socio is computed from the LIHC (Low Income High Cost) energy poverty definition using national statistical office household expenditure data. The R3 modifier amplifies infrastructure risk scores in catchments with high unemployment, elderly concentration, and net outward migration. Coverage status (21 July 2026 audit): V_socio + EP_rate + gdp_per_capita + unemployment_rate are populated for 32/39 countries, but 8 large countries (Italy, Japan, Portugal, Spain, Sweden, Germany, France, US) emit fleet-uniform national aggregates instead of per-substation LAU-2/NUTS-3 values — pending the systemic Wave 4 per-substation interpolation regression fix (sister to R3_C_mult + CMIP6 climate_trajectory). (Task #452 already closed this gap for migration_score specifically via per-substation Niva 2023 raster enrichment; the remaining V_socio / EP_rate / GDP / unemployment fields await Task #450 SYSTEMIC scope.) Catchment population is now populated for all 39/39 countries via the GHSL Population Grid (EC JRC / Copernicus Emergency Management Service, R2023A epoch E2025, Mollweide equal-area) computing 5 km per-substation zonal sums — Task #451 landed 23 July 2026, replacing a legacy synthetic generator with a real GHSL spatial-join pipeline (795k-substation cohort; 0.17% receive Convention #56 visibly-honest None fallbacks for remote/offshore points where the raster has NoData — concentrated in US Aleutians+Pacific, Canada Arctic, Australia outback, Nordic fjords). Migration_score is now populated for all 39/39 countries via the Niva et al. 2023 gridded net-migration dataset (Aalto University / Wittgenstein Centre, published Nature Human Behaviour 7:2023-2037, DOI 10.1038/s41562-023-01689-4, dataset DOI 10.5281/zenodo.7997134, CC BY 4.0 licence) using the 20-yr sum raster 2000-2019 at 5 arc-min (~10 km) resolution — Task #452 landed 23 July 2026, closing the R2 defect via NARROW-scope enrichment (fill Nones) plus fleet-uniform override that retired the 0.5 national scalar fallback of 8 Wave 4 majors (France, Germany, US, Italy, Spain, Portugal, Sweden, Japan, Australia partial, Austria partial) with real per-substation Niva-derived values in [0, 1] range via tanh(x/200) mapping. Genuine pre-Task-#452 per-substation distributions (Norway, Denmark, Lithuania, Ireland, Turkey, Mexico, Greece, New Zealand, Poland partial) are preserved untouched — their semantic-drift issues (e.g. Mexico [-5, +8] percent-scale, Denmark tiny [-0.02, +0.04] scale) are deferred to Task #450 SYSTEMIC scope. Cohort-wide 97.13% real Niva values written + 0.17% Convention #56 visibly-honest None + 2.70% preserved distributions untouched. The remaining 4 countries with major partial-coverage gaps from failed spatial joins (Luxembourg 12% populated / Slovenia 16% / Colombia 51% / Lithuania 50%) are now closed via Task #453 (23 July 2026) using per-substation polygon spatial-join against Eurostat GISCO NUTS-3 2024 shapefile (European Commission / Eurostat, EPSG:4326, annual vintage, open-license attribution-required — for LU/SI/LT) plus GADM 4.1 admin1 shapefile for Colombia (UC Davis geodata.ucdavis.edu/gadm, peer-reviewed academic-open-license derivative of DANE Marco Geoestadístico Nacional — chosen as documented-proxy after DANE geoportal empirically requires form-based interactive download). Task #453 empirical closure: 6,967 v43 substations enriched across LU/SI/LT/CO (LU 634/634 = 100% · SI 1,571/1,574 = 99.8% · LT 4,396/4,396 = 100% · CO 366/366 = 100%; 3 Convention #56 fallback total = 0.04% cohort-wide; 100% preservation of Task #451 catchment_population + Task #452 migration_score audit-trail markers via merge-not-replace BINDING contract). Each Task #453 enriched substation carries per-region gdp_per_capita + unemployment_rate + EP_rate_region + elderly_pct + computed V_socio (via mirrored 0.45·ep_norm + 0.35·gdp_norm + 0.20·elderly_norm formula from socioeconomic.py:511-517 canonical). Task #453 is the ADMIN structural variant within REPORTS_FRAMING_KB.md §8bis Discipline #47 spatial-enrichment family (STOCK = Task #451 GHSL zonal-sum, FLOW = Task #452 Niva point-sampling, ADMIN = Task #453 polygon spatial-join). Discovery motivating Task #453 also codified METHODOLOGY_DISCIPLINES.md §5septies (empirical OSM tag density is per-country) as candidate architectural discipline — Poland + Luxembourg + Slovenia + Colombia + Lithuania empirically ALL show 0.0% populated ref:nuts:3 / addr:department tag rates at country scale, retiring the connector-tag-extraction path as load-bearing for admin-code derivation. Task #454 SYSTEMIC (24 July 2026) extended the Task #453 polygon utility to 11 additional countries with matching v43-at-partial-coverage pattern — 9 EU countries (Belgium + Czechia + Denmark + Estonia + Finland + Ireland + Latvia + Netherlands + Poland) via the same Eurostat GISCO NUTS-3 2024 shapefile already loaded for Task #453, plus 2 non-EU countries via GADM 4.1 admin1 shapefiles (Switzerland cantons + Canada provinces; same UC Davis academic-open-license documented-proxy pattern used for Task #453 Colombia). Two utility refinements landed during Task #454 execution to accommodate new failure modes surfaced empirically: (a) a 3-case skip-logic decision tree with new case-b (province populated by earlier connector pass but socio_economic gap unfilled — Canada case; utility bypasses polygon join and uses the existing province directly as CSV lookup key); (b) Switzerland csv_lookup_aliases with 5 entries mapping GADM English/French canton names (Lucerne, Sankt Gallen etc.) to CSV German canonical forms (Luzern, St. Gallen). Task #454 SYSTEMIC empirical closure: 52,043 v43 substations enriched across 11 new countries (BE 5,428 + CZ 7,825 + DK 2,364 + EE 1,178 + FI 135 + IE 282 + LV 3,425 + NL 3,807 + PL 25,509 + CH 865 + CA 1,227) plus Task #453 4-country idempotent no-op preservation of 6,967 subs = 59,041/59,077 = 99.94% cohort-wide coverage across 15-country combined Task #453+#454 cohort with 65 Convention #56 out-of-polygon fallback (0.11%) and 100% Task #451/#452 marker preservation via merge-not-replace BINDING contract. Wall-clock 15s cumulative (Poland alone 1.3s = ~19,600 subs/sec). Post-Task-#454 Discipline #47 ADMIN variant instance count is 15 (Task #453 4 + Task

#454 11), and Convention #78 §5septies OSM-tag-density empirical instance count is 16 (Poland P21 baseline + Task #453 cohort 4 + Task #454 cohort 11 — well above Convention #76 BINDING promotion threshold of 5-10 empirical instances). All inputs sourced from institutional statistical offices with citable vintage — meeting CSRD Article 29a limited assurance requirements for the 32-country cohort where fields are per-substation, plus Task #453+#454 15-country cohort now closed at Grid Equity axis via polygon spatial-join.

3

Infrastructure Resilience [Re composite home]

FC v3 §14 · SSI v4.2 Re composite (canonical) · EU Taxonomy Art. 11 (mapped)

READY

WHY THIS REPORT EXISTS

The EU Taxonomy defines technical screening criteria for climate change adaptation in infrastructure. This is the lowest-hanging ESG product from the SSI Index — READY for 9 of 10 countries today. The SSI composite score (R_median), 6 components, modifier architecture, and Markov risk outputs provide the quantitative evidence base for Article 11 screening at asset level.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.4 calls for upgrading infrastructure to make it sustainable. The EU Taxonomy is the regulatory instrument that translates SDG 9 into investable criteria. By classifying this substation against Article 11 technical screening criteria — using R_median (0.101), all 6 components, and 5 modifiers — this report provides asset-level Taxonomy alignment that enables sustainable finance reporting.

SDG 9 SDG 13 SDG 12

Substation Profile

R_median (composite)	**0.101**
R_base (pre-modifier)	**0.000**
Modifier Impact	**10050.0%%**
C (Condition)	**—**
V (Voltage)	**—**
I (Insulation/Climate)	**—**
E (Environment)	**—**
S (Seismic)	**—**
T (Transition)	**—**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
`R_median` Composite score	0.101	EU Taxonomy Art. 11: Adaptation screening	READY
`6 Comp.` C/V/I/E/S/T	All present	EU Taxonomy TSC: Physical risk identification	READY
`R3-R7` Modifier suite	All applied	ESRS E1: Adaptation measures evidence	READY
`Markov` Degradation model	—	TCFD: Forward-looking risk assessment	READY
`R5` Asymmetric CI	0.000	ESRS: Uncertainty quantification	GAP

Data Sources & Vintage

ERA5 Climate Reanalysis Copernicus CDS	2024 Weekly
National Seismic Hazard Map National Geological Survey	2023 Multi-year
Economic Statistics (GDP, unemployment) National Statistics Office	2023 Annual
CIGRE TB 761 Markov CIGRE	2019 N/A

TECHNICAL VALIDITY

Article 11 screening uses the full SSI v4.0.2 scoring engine: 6-component weighted composite (C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05), Gaussian copula Monte Carlo (10,000 iterations), and 5 modifiers (R3 social, R4 topology, R5 asymmetric CI, R6a restoration, R7 cyber). The Markov 5-state degradation model provides the forward-looking element. All inputs are from institutional open-source data with citable vintage — meeting CSRD Article 29a limited assurance requirements.

4 Pollution & Corrosion

ESRS E2 Pollution · ISO 9223 Corrosion Classification · Environmental Impact

GAP

WHY THIS REPORT EXISTS

Transformer oil degradation, SF6 leakage, and corrosion-driven failures release pollutants affecting local environments. ESRS E2 requires disclosure of pollution risks. This report assesses the substation's corrosion classification under ISO 9223 and the environmental sensitivity of its surroundings.

SDG Alignment

SDG 11 — Sustainable Cities and Communities (primary)

Target 11.6 aims to reduce the adverse environmental impact of cities. The corrosion class (—) and E2_local enrichment factor (—) track environmental degradation and pollution sensitivity at asset level. Where corrosion is advanced and maintenance deferred, equipment failure risks releasing mineral oil, PCBs (legacy units), and SF6 — directly relevant to urban and peri-urban environmental quality.

SDG 11 SDG 3 SDG 15

Substation Profile

Corrosion Class (ISO 9223)	—
E2 Local Enrichment	—
E Component Score	—
Markov Steady-State	—

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
Corrosion ISO 9223 class	—	ESRS E2: Pollution risk classification	GAP
E2_local Environmental sensitivity	—	ESRS E2: Local environmental impact	GAP
E Environment component	—	ISO 9223: Atmospheric corrosion	GAP

Data Sources & Vintage

Energy Market Data National TSO	2023 Annual
Renewable Installations National DER Registry	2024 Monthly

TECHNICAL VALIDITY

Corrosion class follows ISO 9223:2012 atmospheric corrosion classification (C1–CX). Status depends on whether the corrosion class shows real variance across the fleet or uses default values. Full validation requires national environmental agency air quality monitoring overlay at substation coordinates (SO_2, NO_x, particulate deposition).

5 Energy Transition & DER Stress

GAP

ESRS E1 Transition Plan · TCFD Transition Risk · EU Green Deal Alignment

WHY THIS REPORT EXISTS

> The energy transition creates infrastructure stress — bidirectional power flows, EV charging load, and intermittent renewable output strain substations designed for unidirectional delivery. This report measures whether grid infrastructure is keeping pace with decarbonisation.

SDG Alignment

SDG 7 — Affordable and Clean Energy (primary)

Target 7.2 requires substantially increasing the share of renewable energy. This substation has a DER ratio of —. The T1_score (—) measures the stress this places on the substation, providing the empirical feedback loop between SDG 7 ambition and infrastructure reality.

SDG 7 SDG 13 SDG 9

Substation Profile

T1 Score (transition stress)	—
DER Ratio	—
DER Variability	—
EV Load Ratio	—
T Component Weight	0.05

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
T1 Transition stress score	—	ESRS E1: Transition plan assessment	GAP
DER_ratio Renewable/load ratio	—	TCFD Transition: Technology risk	GAP
DER_var Intermittency	—	EU Green Deal: Grid flexibility	GAP
EV_load EV charging ratio	—	SDG 7: Clean energy access	GAP

Data Sources & Vintage

Air Quality Monitoring National Environmental Agency	2023 Annual
ISO 9223 Corrosion ISO	2012 N/A

TECHNICAL VALIDITY

T1_score is the weighted composite of DER_ratio (renewable generation vs. local demand), DER_variability (intermittency), and EV_load_ratio (electric vehicle charging load as fraction of capacity). DER data sourced from national energy regulators and renewable installation registers. DER_ratio > 1.0 indicates net export during peak generation — a marker of successful energy transition with associated infrastructure stress.

6 Cybersecurity Exposure

NIS2 Directive · ENISA Cybersecurity Index · ESRS G1 Governance

GAP

WHY THIS REPORT EXISTS

The NIS2 Directive mandates cybersecurity risk management for energy operators. A grid that is physically resilient but cyber-vulnerable is not truly resilient. The SSI R7 modifier combines SCADA assessment, communication architecture analysis, and national-level cyber maturity (ENISA/DESI indices) to produce a substation-level digital vulnerability score.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for resilient infrastructure — in digitalised grid systems, this must include cyber resilience. The R7 modifier (—) assesses digital vulnerability combining SCADA exposure, communication protocol security, and national cyber maturity. SDG 16 (Strong Institutions) also applies: NIS2 compliance is a governance imperative, and the betweenness centrality (—) identifies single-point-of-failure risk in the grid topology.

SDG 9 SDG 16

Substation Profile

R7 Cyber Modifier	—
Graph Degree (topology)	—
BC Percentile (centrality)	—
Is Bridge Node	**No**
Cluster Coefficient	—

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R7 Cyber modifier	—	NIS2: Cybersecurity risk management	GAP
BC Betweenness centrality	—	ESRS G1: Governance & topology risk	GAP
Degree Graph degree	—	Grid resilience: Connectivity	GAP

Data Sources & Vintage

Cybersecurity Index
ENISA
2024 Biennial

DESI Connectivity
European Commission
2024 Annual

TECHNICAL VALIDITY

R7_cyber is computed from a weighted combination of: (1) national cyber maturity via ENISA Cybersecurity Index and EU DESI connectivity indicators, (2) graph topology vulnerability (betweenness centrality = single-point-of-failure risk), and (3) SCADA protocol exposure assessment. Substation-level SCADA data is operator-proprietary — the national-level proxy provides a baseline but not asset-specific granularity.

7 SFDR PAI Infrastructure Disclosure

SFDR Article 4 · PAI Table 1 · Delegated Reg (EU) 2022/1288 · FC v3 §14 subsection 13.7

READY

WHY THIS REPORT EXISTS

The SFDR (Sustainable Finance Disclosure Regulation) mandates that financial market participants disclose Principal Adverse Impact (PAI) indicators for infrastructure investments. PAI Table 1 requires quantitative assessment of physical + governance risk at asset level for Article 8/9 fund classification. The SSI v4.2 Re composite (resilience metric per FC v3 §14 subsection 13.7) provides the canonical proxy for Infrastructure PAI: it integrates 6-axis physical risk (C/V/I/E/S/T), 6-modifier degradation chain (R3/R4/R6c/R6d/R6e/R8/R9/R10), and Markov forward-looking trajectory into a single normalised value directly consumable by SFDR reporting.

SDG Alignment

SDG 12 — Responsible Consumption and Production (primary)

Target 12.6 encourages companies to adopt sustainable practices and integrate sustainability information into their reporting cycle. SFDR is the EU regulatory instrument that operationalises this at fund level for infrastructure investors. This substation carries an Re_norm (normalised resilience composite) of 0.493 and Re_raw of 1.347, providing the empirical basis for PAI Table 1 Infrastructure indicator disclosure under Delegated Reg (EU) 2022/1288.

SDG 12 SDG 9 SDG 13

Substation Profile

Re_norm (SFDR PAI composite)	**0.493**
Re_raw (pre-normalisation)	**1.347**
R6c Flood modifier	**1.100**
R6d Wildfire modifier	**1.153**
R6e Winter storm modifier	**1.083**
R8 Adaptation modifier	**0.950**
R9 Compound event modifier	**1.000**
R10 Just-transition modifier	**1.051**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
Re_norm SFDR PAI composite (normalised)	0.493	SFDR Art. 4 · PAI Table 1 Infrastructure	READY
Re_raw Pre-normalisation resilience	1.347	FC v3 §14 subsection 13.7	READY
R6c Flood modifier	1.100	SFDR PAI: Physical acute hazards	READY
R6d Wildfire modifier	1.153	SFDR PAI: Physical acute hazards	READY
R6e Winter storm modifier	1.083	SFDR PAI: Physical acute hazards	READY
R8 Adaptation modifier	0.950	SFDR PAI: Adaptation planning	READY
R9 Compound event modifier	1.000	SFDR PAI: Compound climate risk	READY
R10 Just-transition modifier	1.051	SFDR PAI: Social transition risk	READY

Data Sources & Vintage

No dedicated data sources for this report.

TECHNICAL VALIDITY

Re_norm is the canonical SFDR PAI Infrastructure composite per SSI Index v4.2 methodology (FC v3 §14 subsection 13.7). It integrates the 8 v4.2 modifiers (R6c/R6d/R6e physical acute hazards + R8 adaptation + R9 compound events + R10 just-transition + R7 cyber + R3 social) into a single value normalised against the country-fleet percentile distribution. Convention #56 visibly-honest degradation preserves Re_norm=0.0 + Re_raw=1.0 as neutral defaults for net-new substations awaiting the L2/L3/L4 modifier-chain rescore pass per Convention #78 §4bis.4 two-phase workflow. GAP status reflects sites with fewer than 50% of fleet subs carrying non-default modifiers (typical during a v4.23 L1 refresh window before the follow-on rescore).

7

SFDR Principal Adverse Impact (PAI) Infrastructure v4.2 — Re composite anchor

READY

SFDR Article 7 PAI · ESRS E1-7 climate-resilience disclosure · CSRD double-materiality infrastructure overlay (per FC v3 §14 subsection 13.7)

WHY THIS REPORT EXISTS

SFDR Article 7 requires asset-level disclosure of Principal Adverse Impacts on sustainability factors for infrastructure investments. The Re composite (v4.2 bounded resilience aggregate, bounds [0.920, 1.787] normalised to [0, 1]) is the SSI's canonical driver for PAI Infrastructure exposure — it captures the substation's standing on the new v4.2 resilience-modifier family (R6c flood + R6d wildfire + R6e winter + R8 adapt + R9 compound + R10 just) in a single scalar suitable for SFDR PAI reporting. CSRD double-materiality requires both inside-out (impact on environment) and outside-in (climate-resilience risk to the asset) — Re composite anchors the outside-in dimension for grid infrastructure. This report closes the FC v3 §14 subsection 13.7 R7 SFDR PAI axis on the ESG Radar.

SDG Alignment

SDG 9 — Industry, Innovation, Infrastructure (primary)

Target 9.1 calls for developing quality, reliable, sustainable and resilient infrastructure. The Re composite Re_norm = 0.493 (moderate exposure) at this substation provides the SFDR PAI Infrastructure metric, anchoring climate-resilience disclosure to a single auditable scalar. Re_raw = 1.3470 with methodological bounds [0.920, 1.787] derived from the v4.2 modifier family product range. Lower Re_norm → lower PAI Infrastructure exposure → stronger SFDR Article 7 alignment.

SDG 9 SDG 13 SDG 11

Substation Profile (Re composite)

Substation ID	**AT_v43_80c76c1cf90e**
Re_raw (unbounded)	**1.3470**
Re_norm (bounded [0, 1])	**0.493**
Tier classification	**MODERATE**
Methodology bounds	**[0.920, 1.787]**
Austrian fleet observed range	**1.104 → 1.137**
Coverage	**14,720 substations**

SSI Variables Feeding This Report

SSI VARIABLE	VALUE	ESG DISCLOSURE REQUIREMENT	STATUS
R6c Flood (ISPRA PGRA, ADDITIVE +0.00..+0.30)	+1.100	SFDR PAI 4 (climate physical risk)	READY
R6d Wildfire (EFFIS + FIRMS + SIAB, MULT)	1.153	SFDR PAI 4 (climate physical risk)	READY
R6e Winter / Ice (ERA5 Bourgouin p99, MULT)	1.083	SFDR PAI 4 (climate physical risk)	READY
R8 Adaptive Capacity (E-Control + BMK + DESI Austria)	0.950	ESRS E1-3 Adaptation actions	READY
R9 Compound Hazard (ISPRA RNDA, STEP)	1.000	ESRS E1-9 Anticipated financial effects	READY
R10 Energy Justice (EU-SILC + Austria SILC + national regulator)	1.051	SFDR PAI 14 (board diversity proxy) + ESRS S3	READY
Re_norm aggregate (bounds [0.920, 1.787])	0.493	**SFDR Art. 7 PAI Infrastructure anchor**	READY

Data Sources & Vintage

ISPRA IdroGEO PGRA	2024 (annual)
Copernicus EFFIS + NASA FIRMS	Q4 2025 (active + historical)
ISPRA SIAB national-tier burned-area	CY2024
ECMWF ERA5 Bourgouin freezing-rain p99	1991-2020 reanalysis
E-Control quality regulation indicators + APG smart-meter penetration	2024 annual
ENEA Adaptation Index	2024
DESI Digital Economy index	2024
ISPRA RNDA compound-hazard cross-product	CY2024
EU-SILC ILC_MDES01/03	2024 wave
Eurostat EU-SILC + Statistik Austria EU-SILC (Austrian energy-poverty proxy)	2024
E-Control regional SAIDI	CY2024

TECHNICAL VALIDITY

The Re composite is the bounded aggregate Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c - 1.00) with normalisation Re_norm = clip((Re_raw - 0.920) / (1.787 - 0.920), 0, 1). Bounds [0.920, 1.787] derived from the multiplicative band of the 5 multiplicative v4.2 modifiers at tier extremes plus the R6c additive cliff range. Austrian fleet observed range [0.974, 1.604] across 14,720 substations. SFDR PAI Infrastructure proxy validated against TCFD physical-risk framework and ESRS E1-9 anticipated-financial-effects disclosure. See FC v3 §14 subsection 13.7 for the R7 SFDR PAI axis derivation.

C

Audit Trail & Methodology Note

Traceability for CSRD Article 29a limited assurance

ITEM	VALUE
Scoring Engine	SSI Index **v4.2** — Gaussian copula Monte Carlo (10,000 iterations) · 11 modifiers (5 baseline + 6 v4.2 resilience family) · Re composite bounded [0.920, 1.787]
Weight Architecture	C=0.30, V=0.10, I=0.25, E=0.10, S=0.20, T=0.05
Modifiers Applied	R6c flood: 1.100, R6d wildfire: 1.153, R6e winter: 1.083, R8 adapt: 0.950, R9 compound: 1.000, R10 just: 1.051, R7 cyber_v2: 1.027
Degradation Model	Markov 5-state, CIGRE TB 761 calibration
Thermal Model	IEEE C57.91 transformer loading curves
Report Date	17 March 2026
Reporting Period	Calendar year 2025
Refresh Cadence	Annual (data ingestion → scoring engine → display)
Substation	Station v43_80c76c1cf90e (AT_v43_80c76c1cf90e)
Data Assurance Level	All inputs from institutional open-source with citable vintage
Re composite (v4.2)	Re_raw = 1.3470 · Re_norm = 0.493 · MODERATE exposure · bounds [0.920, 1.787]

D

Re composite — bounded resilience aggregate (v4.2)

Resilience composite anchoring CSRD/ESRS climate-resilience disclosure

The Re composite (v4.2) is a bounded resilience aggregate capturing the new R6c flood / R6d wildfire / R6e winter / R8 adapt / R9 compound / R10 just modifier family in a single scalar suitable for CSRD/ESRS climate-resilience disclosure:

```
Re_raw = (R6d × R6e × R8 × R9 × R10) + (R6c − 1.00)
```

Bounds: **[0.920, 1.787]**. Re_raw < 1.000 when R8_adapt < 1.00 dominates other near-neutral factors. Austrian fleet observed range across 14,720 substations: the methodology-bounded extremes [0.920, 1.787] — low-end values arise at interior low-exposure substations with R8_adapt < 1.00 dominating near-neutral other modifiers, high-end values at coastal compound-exposure substations where R6c flood + R9 compound stack multiplicatively. v4.2 explicitly retires the pre-Path-B "winter exposure is a real climate burden" reading per Convention #6 §2.3.1 (Bourgouin freezing-rain p99 tail-event amplifier per §6 design — winter exposure signal correctly lives in v4.0.2 I1 Snow/Ice metric, where it methodologically belongs). See W1-W10 FC v2 for per-substation worked examples + W4 Climate resilience baselines.

Foundation contact

The SSI Index is a non-profit foundation project, please mail to ssi_index@ikenga.eu for more details.